PS#8: Improper integrals - Answer key

1. (Activity 6.5.4) For each of the following definite integrals, decide whether the integral is improper or not. If the integral is proper, evaluate it using the First FTC. If the integral is improper, determine whether or not the integral converges or diverges; if the integral converges, find its exact value.

   (a) $\int_0^1 \frac{1}{x^{1/3}}\,dx$ – **improper**, **converges** to $\frac{3}{2}$

   (b) $\int_0^2 e^{-x}\,dx$ – **proper**, **converges** to $-e^{-2}+1$

   (c) $\int_1^4 \frac{1}{\sqrt{4-x}}\,dx$ – **improper**, **converges** to $2\sqrt{3}$

   (d) $\int_{-2}^2 \frac{1}{x^2}\,dx$ – **improper**, **diverges**

   (e) $\int_0^{\pi/2} \tan(x)\,dx$ – **improper**, **diverges**

   (f) $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$ – **improper**, **converges** to $\frac{\pi}{2}$